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RESEARCH ARTICLE

# Genome-wide Identification, Expression Profiling and Promoter Analysis of Trehalose-6-Phosphate Phosphatase Gene Family in Rice

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## Abstract
Trehalose-6-phosphate phosphatase (TPP) plays a key role in trehalose metabolism in plants. Here, we performed comprehensive in silico analyses and identified 12 OsTPPs (*Oryza sativa* TPPs) utilizing various bioinformatics tools. Phylogenetic tree, accomplished with OsTPPs and TPPs from 11 monocot and dicot species, was divided mainly into two clades, each clade containing six OsTPPs. Exon–intron distribution was related to phylogenetic clades. All OsTPPs are distributed within nine chromosomes (chr.), except Chr. 1, Chr. 5 and Chr. 11. OsTPPs were found to be stable in nature according to the 3-D structure prediction. *Cis*-regulatory elements (CREs) were also analyzed using 2 kb upstream of start codon for each gene to predict their biological functions. We categorized all CREs in five distinct groups based on core elements, stress response, cellular development, hormonal regulation, and unknown function, distributed in a range of 3–14 CREs in each group. Interestingly, our expression analysis showed that OsTPPs were more upregulated in response to drought and cold stresses compared to salt stress. Abundance of stress-related CREs found signifies TPPs' possible role in stress response, which may facilitate to find related transcription factors and unveil complex molecular mechanisms during stress response.

**Keywords** *Cis*-regulatory elements · Gene expression · In silico · Rice · Trehalose-6-phosphate phosphatase

## Introduction

Trehalose is a non-reducing disaccharide composed of two molecules of α-glucose, widely distributed in yeast, bacteria, fungi, animals, insects and plants (Elbein et al. 2003; Lunn et al. 2014). Previously, trehalose was thought to be present in only a few desiccation-tolerant plants (Bianchi et al. 1991; Drennan et al. 1993; Albini et al. 1994), but eventually discovered in *Arabidopsis* (*Arabidopsis thaliana*) with the identification of two enzymes, namely, Trehalose-6-phosphate synthase (TPS) (Blázquez et al. 1998) and Trehalose-6-phosphate phosphatase (TPP) (Vogel et al. 1998). Trehalose biosynthesis in plants as well as in other organisms, except the vertebrates, occurs in two main steps incorporating two catalytic enzymes TPS and TPP. In the first step, Trehalose-6-phosphate (T6P), a phosphorylated intermediate, is synthesized from Uridine diphosphate-glucose (UDPG) and Glucose-6-phosphate (G6P) by TPS. In the second step, T6P is dephosphorylated by TPP to synthesize trehalose. Later, trehalose is hydrolyzed by trehalase (TRE1) to produce two molecules of glucose (Avonce et al. 2006; Ponnu et al. 2011). Thus, trehalose biosynthesis involves three enzymes, namely TPS, TPP and TRE1, but only TRE1 is encoded by a single copy of gene, whereas TPS and TPP contain multigenic families to be encoded (Lunn 2007).

Trehalose functions in plants diversely and confers its essential role in different stages of development, such as embryo formation and flowering (Iturriaga et al. 2009). Trehalose can accumulate in plants without showing any adverse phenotypic effects (Shim et al. 2019). Trehalose synthesis is found to be related to storage carbohydrate synthesis. In *Arabidopsis*, exogenous application of trehalose induced the large subunit of Adenosine diphosphate (ADP)-glucose pyrophosphorylase *Apl3* expression (Wingler et al. 2000). Trehalose induced fructan accumulation in *Hordeum*

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*vulgare* in the presence of glucose or mannitol by increasing the activities of Suc:Suc-1-fructosyltransferase and Suc:fructan-6-fructosyltransferase enzymes (Wagner et al. 1986; Müller et al. 2000). Importance of trehalose metabolism in photosynthesis was also observed. For instance, transformed tobacco plant with *Escherichia coli* TPS gene *ostA* resulted in increase of photosynthesis per unit leaf area through enhancing RUBISCO content, but a TPP gene *ostB* showed the opposite impact (Pellny et al. 2004). The role of trehalose in favor of plants against different biotic and abiotic stress conditions is also important. For instance, effects of trehalose upon stresses have been studied and, thus, being used for engineering plants with better response against those stresses. Trehalose genes in *Zygophyllum xanthoxylum* (*ZxTPP*) facilitate the osmotic adjustment process to increase tolerance during drought stress (Wu and Su 2016). Transgenic tobacco plants transformed with *E. coli ostA* and *ostB* were better tolerant against water scarce or drought condition (Pilon-Smits et al. 1998). Also, transgenic tobacco plants expressing *E. coli ostA* were better tolerant against salt stress, and transgenic seeds showed improved germination efficiency possibly due to increased accumulation of trehalose to protect the developing young embryos (Jun et al. 2005). Similarly, rice plants transformed with *ostA* and *ostB* showed higher trehalose synthesis, and performed better under salt, drought, cold stresses compared to the non-transgenic plants (Garg et al. 2002). Tomato and *Arabidopsis*, transformed with overexpressing trehalose genes showed resistance against abiotic stresses (Cortina and Culiáñez-Macià 2005; Miranda et al. 2007). Trehalose also confers important roles against pathogens (Farías-Rodríguez et al. 1998; Streeter and Gomez 2006; Suárez et al. 2008).

Alteration in the dephosphorylation of T6P by TPP during trehalose metabolism can consequently modify trehalose amount, which affects the normal growth and also responses under stress conditions. It has been reported that spermidine is an antagonist of trehalose and seed priming with both results in enhanced chilling tolerance in rice (Fu et al. 2020). Pretreated tobacco leaf with trehalose showed enhanced resistance against tobacco mosaic virus. A signal transduction pathway involving calcium and Reactive Oxygen Species (ROS) was induced by exogenous trehalose, which could improve abiotic stress response as well (Shi et al. 2019). Overexpression of *OsTPP1* and *OsTPP3* activated stress responses genes, which conferred drought resistance in rice and maize (Ge et al. 2008; Jiang et al. 2019). *AtTPPI* knockout mutant showed multiple phenotypic expressions, such as smaller leaves, shortened roots, delayed flowering, sucrose dependent seedling development, etc. AtTPPI was multi-targeted to peroxisomes, nucleus, plastids, and nucleolus because of conserving peroxisomal targeting signal type 1 tripeptides (Kataya et al. 2020). In maize, Ra3/GRMZM2G151044 is localized in nucleus and cytoplasm. Ra3/GRMZM2G151044 and TPP4 mutants reduced meristem development and increased inflorescence branching. Complementation of *ra3* by a catalytically inactive version of RA3 confirms that a non-enzymatic function plays the role in meristem determinacy (Claeys et al. 2019). *OsTPP1* and *OsTPP2* showed up-regulated expression under different types of stress conditions like drought, cold and salt, and also after ABA treatment (Pramanik and Imai 2005; Shima et al. 2007). *OsTPP1* expression was continuously found to be higher than *OsTPP2* in various abiotic stress conditions (Shima et al. 2007). Transient expression of *OsTPP1* in chilling stress changed the levels of glucose and fructose (Pramanik and Imai 2005). OsMAPK3 phosphorylated transcription factor OsICE1 led to accumulation of phospho-OsHOS1 and trehalose by increasing the *OsTPP1* expression, consequently enhanced chilling tolerance in rice (Zhang et al. 2017). Transmutation activity of TPP was consistent with the shifted transcript level of the corresponding gene and trehalose level. Infective larvae of *Anisakis simplex* accumulated trehalose and glycogen during the thermal stress under starvation to provide energy through the molecular and biochemical regulation of trehalose and glycogen metabolism (Łopienska-Biernat et al. 2019). Susceptible seedlings of maize showed reduced *ZmTPP1* expression, but resistant seedlings showed enhanced expression after drought stress. Moreover, ZmTPP1 showed significant similarities with rice TPPs (Acosta-Pérez et al. 2020). *Cis*-regulatory elements (CREs) of promoters of TPP act to induce trehalose metabolism and better response against stress conditions. *AtTPPI* expression was directly regulated by ABA-responsive elements like ABF1, ABF2, and ABF4 to enhance drought tolerance by adjusting stomatal apertures in *Arabidopsis* (Lin et al. 2020). In the presence of ABA, the ABA-responsive transcription factor ABF2 directly bound with *AtTPPE* promoter to activate its expression for root elongation and stomatal movement by inducing ROS accumulation (Wang et al. 2020). TPP enzymes can act upstream of the RA1 transcription factor to regulate inflorescence branching and might play as an important signal (Acosta-Pérez et al. 2020). AtTPPF over-expressers showed reduced $H_2O_2$ accumulation under drought stress. In plants undergoing drought stress, *AtTPPF* transcription was probably up-regulated by DREB1A as it bound to the DRE/CR motif in *AtTPPF* promoter (Lin et al. 2019).

Rice is a major cereal crop feeding more than half of the population worldwide, which has been cultivated for more than 7000 years, one of the oldest domesticated crops by human (Izawa and Shimamoto 1996). This particular crop bestows us with providing the cheapest carbohydrate source, and is considered as an excellent model of studying plant biology after *Arabidopsis*, for its complete genome sequencing and availability of sufficient mutants. Numerous studies have been done and still are ongoing focusing on the improvement of rice yield and enhancing its tolerance against various environmental stress conditions. Here, we have studied rice TPP genes in silico by identification of *OsTPP*s, gene duplication analysis, phylogenetic tree analysis with 11 more plant species, motif and domain analyses, sub-cellular localization prediction, 3-D structure modeling of proteins, investigation of CREs, and analyzing gene expression, to contribute to the better understanding of OsTPP functions.

## Materials and Methods

### Identification of TPP Genes and Their Chromosomal Location

To conduct genome-wide identification of TPP genes, first, we collected the information of genes from GreenPhyl v4 (https://www.greenphyl.org/cgi-bin/index.cgi) by searching one known gene *OsTPP1* (*LOC_Os10g44230*) (Ge et al. 2008). Then, we retrieved the sequences of genomic DNA and coding DNA (CDS), and protein from rice genome annotation project web tool (https://rice.plantbiology.msu.edu/analyses_search_locus.shtml). Ensembl Plants database (https://plants.ensembl.org/Oryza_sativa/Info/Index) was used to identify Open Reading Frame (ORF) of TPP genes. ProtParam tool (https://web.expasy.org/protparam/) was used to find the length, molecular weight, iso-electric point (pI) and grand average of hydropathicity (GRAVY) of TPP proteins. Chromosomal locations of the genes were mapped using the Shigen Map Tool (https://viewer.shigen.info/oryzavw/maptool/MapTool.do) and subsequently modified using Adobe Illustrator CS6.

### Gene Duplication and Calculations of Ka/Ks Ratios

To check gene duplication, an NCBI BLAST search was performed based on the percentage of query cover to identity of the OsTPP genes against each other (Johnson et al. 2008). Duplicated gene pairs were found out based on BLAST search and phylogenetic tree, subsequently, duplication events were calculated from synonymous rate (Ks), non-synonymous rate (Ka) and evolutionary constraint (Ka/Ks ratio) using Clustal Omega for aligning the protein sequences of the gene pairs (Sievers and Higgins 2014). We then used PAL2NAL web tool for converting to codon alignments of aligned protein and corresponding cDNA sequences (Suyama et al. 2006). The resulted codon alignment was used to calculate Ka and Ks using the CODEML program of PAML (Yang 2007). Ks was reported to represent time frame values, later, the values were translated into divergence time in millions of years using the equation $T = \text{Ks}/2\lambda$, assuming $1.5 \times 10^{-8}$ substitutions/synonymous site/year as the clock-like rates (λ) of synonymous substitution (Koch et al. 2000).

### Phylogenetic Analysis, Exon–Intron Distribution and 3-D Structure Modelling

Full-length protein sequences of 12 species were aligned using MUSCLE2 (Codon) in MEGA X (Kumar et al. 2018). After the alignment, a phylogenetic tree was constructed using MEGA X with the following parameters: Neighbor-Joining method (Nei and Saitou 1987), JTT matrix-based method (Jones et al. 1992), Poisson correction, pair-wise deletion and bootstrap values in percentages with 1000 replicates (Felsenstein 1985). The CDSs and genomic sequences of OsTPP genes were aligned with the Gene Structure Display Server web tool (https://gsds.cbi.pku.edu.cn/) to analyze their exon–intron distribution (Hu et al. 2015). 3-D structure analyses of OsTPPs were done using SWISS-MODEL Workspace web tools (https://swissmodel.expasy.org/interactive#sequence), GASS-WEB, SOPMA secondary structural method (https://npsa-prabi.ibcp.fr/cgi-bin/npsa_automat.pl?page=/NPSA/npsa_sopma.html) and RAMPAGE server (https://mordred.bioc.cam.ac.uk/~rapper/rampage.php) (Combet et al. 2000; Lovell et al. 2003; Izidoro et al. 2015; Moraes et al. 2017; Waterhouse et al. 2018).

### Protein Motif and Domain Analysis and Sub-cellular Localization Prediction

To analyze the motifs of OsTPPs, we used MEME suite 5.1.1 (Bailey et al. 2009). Maximum number of motifs to find was set to 9, minimum width 6, maximum width 50, and any number of repetitions were chosen for the site distribution. The TPP domain [trehalose-phosphatase (Trehalose_PPase); PF02358] was obtained from Pfam database (Finn et al. 2016) and the structures were drawn using DOG Illustrator (Ren et al. 2009). Sub-cellular localizations were predicted using Wolfpsort (https://wolfpsort.hgc.jp/).

### *Cis*-Regulatory Elements (CREs) and Publicly Available Expression Data Analyses

To identify the CREs, first, we collected promoter sequences (2000 bp upstream of start codon) of 12 OsTPPs from NCBI database (https://www.ncbi.nlm.nih.gov/) and then submitted those sequences to PlantCARE (Lescot 2002). Statistical analysis and data visualization were done by Netbeans IDE 8.0 and subsequently edited in Photoshop CS6 (NetBeans 2015). Gene expression was analyzed from Genevestigator RNAseq public anatomy data (Hruz et al. 2008). Then, expression was visualized using MeV tool.

### Quantitative Real-Time Reverse Transcription PCR Analysis

A *japonica* cultivar *Dongjin* was used for expression analysis. Seeds were germinated on half-MS and then transferred to soil in greenhouse under a 14 h–10 h light–dark period at a 24–28 ℃ and 60–70% relative humidity. 6-week-old plants were pulled out carefully and roots were washed with clean water. After that, plants were kept on liquid half-MS and kept inside a growth chamber for acclimatization under the same condition mentioned for greenhouse. 20% PEG-6000 and 150 mM NaCl were mixed with liquid half-MS to create drought and salt stress conditions, respectively. In case of cold stress, plants on liquid half-MS were kept inside a refrigerator at 8 ℃. Leaf samples were collected from the plants subjected to stress treatment for total RNA extraction at 0 h (as control), 3 h, 6 h, 12 h and 30 h.

Qiagen RNeasy mini kit was used for leaf total RNA extraction according to the manufacturer's instructions (QIAGEN). 2 µg of total RNA was used for cDNA generation using ReverTra Ace®qPCR RT Master Mix with gDNA Remover (Toyobo, Osaka, Japan). Quantitative real-time reverse transcription PCR (qRT-PCR) was performed using Prime Q-Master Mix 2X (GENTBIO) on a Step-One Plus Real-Time PCR system (Applied Biosystems). Transcript levels of target genes were normalized with the rice *Ubiquitin5* (*OsUbi5; LOC_Os01g22490*) (Jain et al. 2006). The primer sequences used for qRT-PCR analyses are listed in Supplementary Table 1.

## Results

### Identification of Rice TPP Genes

We identified 12 OsTPP genes in rice named *OsTPP1* to *OsTPP12* (Table 1). ORFs of these genes ranged from 771 bp (*OsTPP12*) to 4,014 bp (*OsTPP10*) and protein length varied from 256 aa (OsTPP12) to 500 aa (OsTPP8) (Table 1). Molecular weight varied from 29.10 KDa (OsTPP11) to 53.44 KDa (OsTPP8) (Table 1). According to the predicted pI value, 8 genes (*OsTPP1*, *2*, *3*, *8*, *9*, *10*, *11* and *12*) were found acidic (< 7) and 4 genes (*OsTPP4*, *5*, *6* and *7*) were found basic (> 7) (Table 1). GRAVY of the OsTPPs ranged from -0.332 to 0.154. All of the OsTPPs showed negative GRAVY values except OsTPP8 (0.154) (Table 1). Negative GRAVY value indicates that the protein is non-polar, hydrophilic in nature and positive value indicates that the protein is polar, hydrophobic in nature (Bhattacharya et al. 2018) (Table 1). Aliphatic Index and Instability Index were also calculated. The Aliphatic Index is the relative volume occupied by Alanine, Isoleucine, Leucine and Valine amino acids having aliphatic side chains (Bhattacharya et al. 2018). Aliphatic Index ranges observed were 77.14 (OsTPP4) to 94.06 (OsTPP12), and Instability Index ranges were 28.77 (OsTPP3) to 48.88 (OsTPP1) (Supplementary Table 6). Higher Aliphatic Index of a protein sequence indicates that the protein may function in wide range of temperatures, and Instability Index indicates whether a protein would be found as stable or unstable condition (Yaqoob et al. 2016).

### Chromosomal Location and Gene Duplication

By mapping 12 OsTPP genes on the rice chromosomes (chr.), we found that physical locations were distributed unevenly on 9 out of 12 chromosomes. Paired genes, named *OsTPP1* and *OsTPP4* located on chr. 2, *OsTPP3* and *OsTPP10* on chr. 7, and *OsTPP6* and *OsTPP12* on chr. 8. Single genes *OsTPP2*, *OsTPP5*, *OsTPP7*, *OsTPP8*, *OsTPP9* and *OsTPP11* were located on chr. 10, chr. 4, chr. 9, chr. 6, chr. 3 and chr. 12, respectively (Fig. 1). No gene was found in chr. 1, chr. 5 and chr. 11 (Fig. 1). We also performed BLAST search of all the genes against each other to find out any presence of gene duplication using National Center for Biotechnology Information (NCBI) web BLAST tool. Genes are considered as duplicated when the query cover and identity value is ≥ 80% (Kong et al. 2013). It is also reported that protein sequence identity of > 70% and the sequence length identity of > 75% of genes are considered as duplicated genes (Gu et al. 2002). Sequence identity among the 12 OsTPP proteins ranged from 31.86% (between OsTPP7 and OsTPP12) to 76.96% (between OsTPP6 and OsTPP7) (Supplementary Table 2). Two genes, separated by less than five genes in a 100 kb region, are called tandemly duplicated genes (Shi et al. 2019). Accordingly, we did not find any tandemly duplicated genes. Paralogs were considered for three pairs of inter-chromosomal duplicated genes, *OsTPP3-OsTPP9*, *OsTPP4-OsTPP8* and *OsTPP6-OsTPP7* (Fig. 1 and Supplementary Table 3). All duplicated gene pairs were found in the same clade of phylogenetic tree (Fig. 2a). Thereafter, we estimated Ka/Ks ratio of these duplicated genes and found that ratios were less than 1, indicating that they have been evolved by functional constraint with negative or purifying selection (Hurst 2002) (Supplementary Table 3). The range of divergence time was 45 to 1,436 million years ago (MYA), indicating that *OsTPP4* and *OsTPP8* gene pairs had recently been duplicated (45 MYA) (Supplementary Table 3).

### Exon–Intron Distribution

Among the OsTPPs, three genes, *OsTPP3*, *OsTPP4* and *OsTPP8*, each of them has 10 introns; five genes, *OsTPP1*, *OsTPP2*, *OsTPP5*, *OsTPP6* and *OsTPP10*, each of them has 9 introns; two genes, *OsTPP11* and *OsTPP12*, each has 7 introns; *OsTPP7* has 6 introns; and *OsTPP9* has 4 introns

**Table 1** Detailed information about the OsTPP genes and corresponding proteins in *Oryza sativa*

| Gene name | Gene ID | PC | ORF (bp) | Chromosome location | | | | | Intron | Length (aa) | M.W. (KDa) | pI | GRAVY | SLoc |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | No | St | Start | End | Length | | | | | | |
| *OsTPP1* | LOC_Os02g44230.1 | IIA | 1782 | 2 | – | 26,767,603 | 26,771,633 | 35,937,250 | 9 | 371 | 41.13 | 5.95 | − 0.160 | Nuclear |
| | LOC_Os02g44230.2 | – | 1751 | 2 | – | 26,767,603 | 26,771,609 | 35,937,250 | 9 | 371 | 41.13 | 5.95 | − 0.160 | Nuclear |
| | LOC_Os02g44230.3 | – | 1738 | 2 | – | 26,767,603 | 26,771,633 | 35,937,250 | 9 | 371 | 41.13 | 5.95 | − 0.160 | Nuclear |
| | LOC_Os02g44230.4 | – | 1833 | 2 | – | 26,767,603 | 26,771,633 | 35,937,250 | 5 | 291 | 31.76 | 8.33 | 0.023 | Nuclear |
| | LOC_Os02g44235.1 | – | 1321 | 2 | – | 26,767,603 | 26,770,876 | 35,937,250 | 8 | 324 | 36.10 | 6.98 | − 0.183 | Chloroplast |
| *OsTPP2* | LOC_Os10g40550.1 | IIB | 2027 | 10 | – | 21,706,909 | 21,710,977 | 23,207,287 | 9 | 382 | 42.58 | 5.84 | − 0.291 | Nuclear |
| | LOC_Os10g40550.2 | – | 1850 | 10 | – | 21,706,982 | 21,710,973 | 23,207,287 | 9 | 382 | 42.58 | 5.84 | − 0.291 | Nuclear |
| | LOC_Os10g40555.1 | – | 1077 | 10 | – | 21,707,372 | 21,709,277 | 23,207,287 | 9 | 358 | 40.12 | 5.83 | − 0.332 | Chloroplast |
| *OsTPP3* | LOC_Os07g43160.1 | IA | 1518 | 7 | + | 25,867,212 | 25,871,233 | 29,697,621 | 10 | 366 | 40.00 | 6.57 | − 0.180 | Cytoplasmic |
| *OsTPP4* | LOC_Os02g51680.1 | IA | 1423 | 2 | + | 31,663,080 | 31,665,818 | 35,937,250 | 10 | 367 | 40.02 | 8.88 | − 0.235 | Cytoplasmic |
| *OsTPP5* | LOC_Os04g46760.1 | IIA | 1050 | 4 | - | 27,726,325 | 27,728,491 | 35,502,694 | 10 | 349 | 39.01 | 7.53 | − 0.160 | Chloroplast |
| *OsTPP6* | LOC_Os08g31630.1 | IB | 2698 | 8 | + | 19,578,770 | 19,583,372 | 28,443,022 | 9 | 370 | 41.00 | 9.29 | − 0.198 | Chloroplast |
| | LOC_Os08g31630.2 | – | 2891 | 8 | + | 19,578,779 | 19,583,372 | 28,443,022 | 6 | 297 | 32.72 | 9.45 | 0.008 | Chloroplast |
| *OsTPP7* | LOC_Os09g20390.1 | IB | 1575 | 9 | + | 12,251,875 | 12,254,087 | 23,012,720 | 6 | 375 | 41.12 | 9.12 | − 0.147 | Chloroplast |
| | LOC_Os09g20390.2 | – | 1719 | 9 | + | 12,251,936 | 12,254,078 | 23,012,720 | 4 | 330 | 36.02 | 8.60 | − 0.132 | Chloroplast |
| | LOC_Os09g20390.3 | – | 1711 | 9 | + | 12,251,891 | 12,254,078 | 23,012,720 | 2 | 260 | 27.68 | 8.72 | − 0.083 | Chloroplast |
| *OsTPP8* | LOC_Os06g11840.1 | IA | 1638 | 6 | – | 6,277,593 | 6,281,865 | 31,248,787 | 10 | 500 | 53.44 | 5.98 | 0.154 | Plasma Membrane |
| *OsTPP9* | LOC_Os03g26910.1 | IA | 1698 | 3 | + | 15,381,803 | 15,384,274 | 36,413,819 | 4 | 374 | 40.21 | 5.72 | − 0.150 | Cytoplasmic |
| *OsTPP10* | LOC_Os07g30160.1 | IIB | 4014 | 7 | + | 17,816,192 | 17,821,232 | 29,697,621 | 9 | 359 | 40.44 | 6.8 | − 0.317 | Cytoplasmic |
| *OsTPP11* | LOC_Os12g09060.1 | IIA | 786 | 12 | – | 4,732,014 | 4,735,251 | 27,531,856 | 7 | 261 | 29.10 | 5.44 | − 0.173 | Cytoplasmic |
| *OsTPP12* | LOC_Os08g25410.1 | IIA | 771 | 8 | – | 15,456,565 | 15,461,336 | 28,443,022 | 7 | 256 | 29.29 | 5.51 | − 0.332 | Chloroplast |

*PC* Phylogenetic Clade, *ORF* Open Reading Frame, *bp* base pair, *No* Number, *St*, Strand, *M.W.* Molecular Weight, *aa* Amino Acid, *KDa* Kilo Dalton, *pI* Iso-electric Point, *GRAVY* Grand average of hydropathy, *SLoc* Sub-cellular Localizations

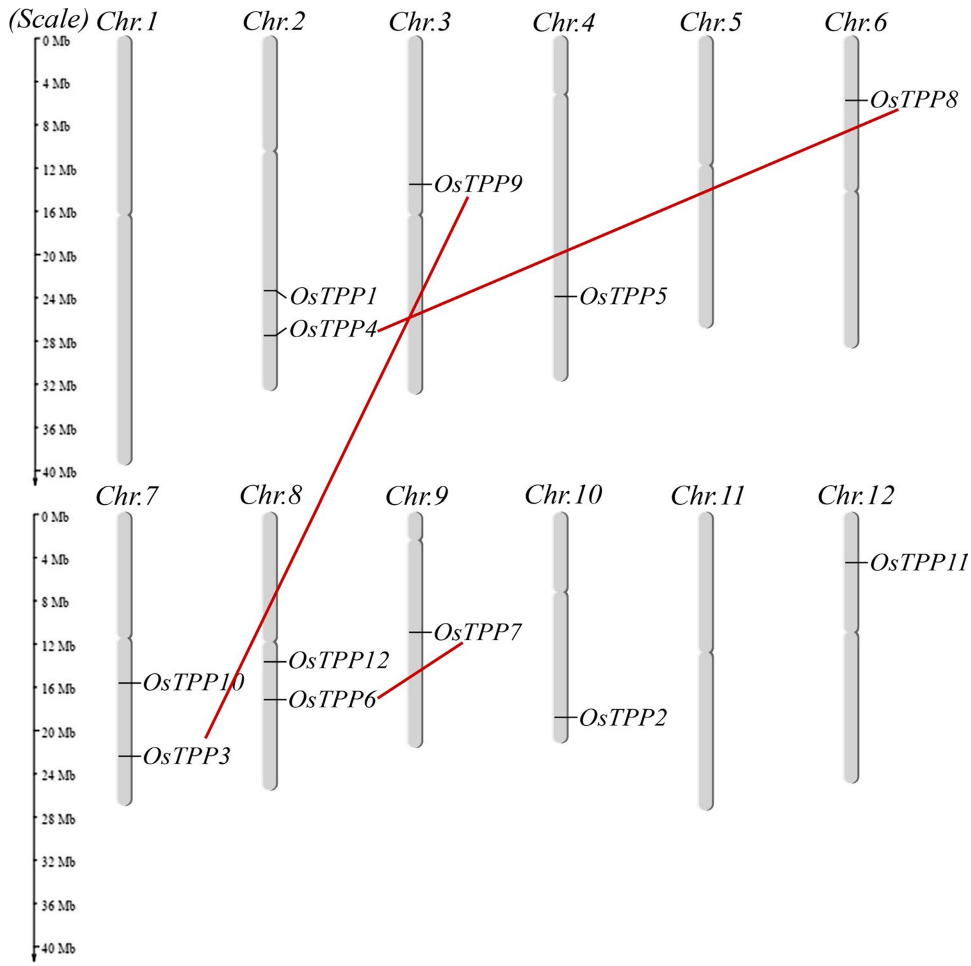



**Fig. 1** Chromosomal location and gene duplication of *OsTPP*s. Chromosome (Chr.) numbers noted above indicate each individual chromosome, and red lines represent the duplicated gene pairs. Chromosome sizes and gene positions were estimated using the scale in megabases (Mb) to the left of the Figure

(Fig. 3a). Moreover, only four genes, namely *OsTPP1, OsTPP2, OsTPP9* and *OsTPP10*, showed phase 0 (intron does not disrupt a codon); seven genes, namely *OsTPP3, OsTPP4, OsTPP6, OsTPP7, OsTPP8, OsTPP11* and *OsTPP12*, showed phase 1 (intron disrupts a codon between the first and second bases); two genes, namely *OsTPP5* and *OsTPP12*, showed phase 2 (intron disrupts a codon between the second and third bases); and only *OsTPP12* showed all of the phases (Phase 0,1,2) (Fig. 3a).

## Sub-cellular Localizations

Sub-cellular localizations were predicted using Wolfpsort (https://wolfpsort.hgc.jp/). Sub-cellular localizations of the OsTPP proteins were predicted in cytoplasm, nucleus, chloroplast and plasma membrane (Table 1). OsTPP1 and OsTPP2 showed nuclear localization. OsTPP3, OsTPP4, OsTPP9, OsTPP10 and OsTPP11 showed cytoplasmic localization. OsTPP5, OsTPP6, OsTPP7 and OsTPP12 showed chloroplast localization. Only OsTPP8 showed plasma membrane localization prediction (Table 1).

## Phylogenetic Analysis

Phylogenetic analysis of OsTPP gene family was done previously with few species. It has been reported that *Arabidopsis*, maize, purple false brome and poplar have 10, 11, 10 and 8 TPP genes, respectively (Shima et al. 2007; Ge et al. 2008; Henry et al. 2014). Those studies also reported rice TPPs, but the numbers were inconsistent. Therefore, in this study, we identified 12 OsTPPs and characterized them in silico (Shima et al. 2007; Ge et al. 2008; Henry et al. 2014). To explore more about OsTPP evolutionary factors, we analyzed phylogenetic tree with 11 more species (Fig. 2a and Supplementary Table 4). To construct phylogenetic tree, we aligned total 108 protein sequences of 12 species, five species from monocot (49 proteins): *Oryza sativa* (12), *Zea maize* (11), *Hordeum vulgare* (6), *Sorghum bicolor* (10), *Brachypodium distachyon* (10); and 7 species from dicot (59 proteins): *Arabidopsis thaliana* (10), *Vitis vinifera* (7), *Carica papaya* (3), *Populus trichocarpa* (8), *Glycine max* (17), *Solanum lycopersicum* (7), *Solanum tuberosum* (7). Phylogenetic tree was mainly divided into two clades, Clade I and Clade II. Clade I was further divided into four sub-clades,

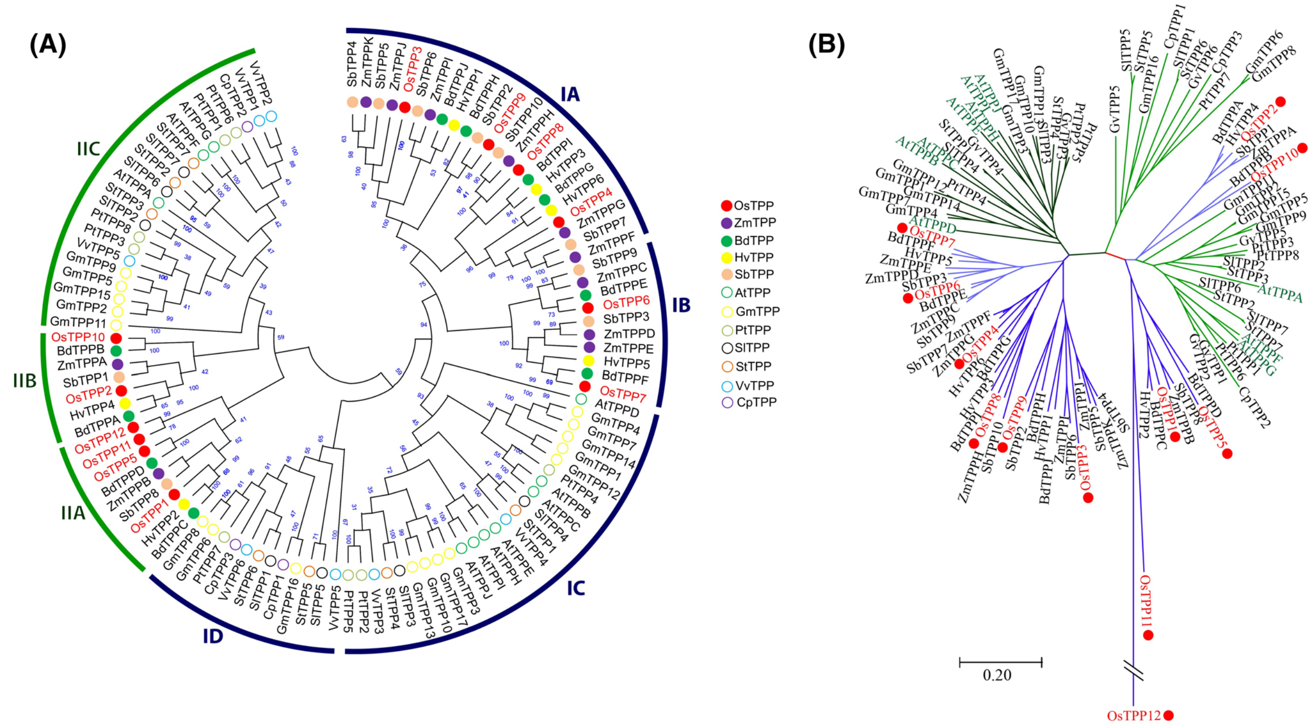



**Fig. 2** Phylogenetic tree and radiation tree of TPPs. **a** Phylogenetic tree of OsTPPs along with maize, barley, sorghum, false brome, arabidopsis, grapevine, papaya, black cottonwood, soybean, grape vine, tomato and potato. Full-length polypeptide sequences were used to make the phylogenetic tree. The sequences were aligned using MEGA X MUSCLE (Codon). The tree was constructed by Neighbor-Joining method in MEGA X. The numbers at the branches represent the percent bootstrap values based on 1000 replications. Monocotyledons and dicotyledons are represented as solid and normal circles, respectively. Therefore, species acronym was added before each TPP protein name. *Os Oryza sativa* (Rice), *Zm Zea mays* (Maize), *Hv Hordeum vulgare* (Barley), *Sb Sorghum bicolor* (Sorghum), *Bd Brachypodium distachyon* (Purple false brome), *At Arabidopsis thaliana* (Arabidopsis), Vv *Vitis vinifera* (Grapevine), *Cp Carica papaya* (Papaya), *Pt Populus trichocarpa* (Black cottonwood), *Gm Glycine max* (Soybean), *Sl Solanum lycopersicum* (Tomato), *St Solanum tuberosum* (Potato). **b** Radiation tree of OsTPPs along with maize, barley, sorghum, purple false brome, arabidopsis, grapevine, papaya, black cottonwood, soybean, grape vine, tomato and potato

namely clade IA, clade IB, clade IC and clade ID; and clade II into three sub-clades, namely clade IIA, clade IIB and clade IIC. Clade I contained 70 protein members (64.8% of total proteins), whereas clade II contained only 38 protein members (35.2% of total proteins). Dicot and monocot plants were well grouped in each clade. Clade IA, clade IB, clade IIA and clade IIB belonged to monocot plants, whereas clade IC, clade ID and clade IIC belonged to dicot plants. Twelve OsTPP proteins were positioned into each main clade equally. Clade IA and clade IB contained four (OsTPP3, 4, 8, 9) and two (OsTPP6, 7) proteins, respectively. Clade IIA and clade IIB contained four (OsTPP1, 5, 11, 12) and two (OsTPP2, 10) proteins, respectively (Fig. 2a).

## Protein Motif and Domain Analyses

We analyzed motif sequences for 12 OsTPPs using MEME suite 5.1.1 and identified nine significant motifs (motif 1–9) (Fig. 3c). Motif 2 and motif 7 were found to be conserved in all OsTPPs. Nearly all the motifs, except motif 8 were highly conserved in first ten OsTPPs (OsTPP1 to OsTPP10). OsTPP11 contained only four motifs (motif 1, motif 2, motif 4 and motif 7). The lowest number of motifs was observed in OsTPP12, containing only three motifs (motif 2, motif 7 and motif 8) (Fig. 3c). To identify the significant component domains, we used Pfam database (Finn et al. 2014). Specific Trehalose_PPase domain (PF02358) was found in all the OsTPP proteins. But, Trehalose_PPase domain found in OsTPP11 and OsTPP12 was in partial or truncated form (Fig. 3b). Moreover, OsTPP3, OsTPP4, OsTPP5, OsTPP6, OsTPP7, OsTPP8, OsTPP9 and OsTPP12 had Low Complexity (LC) region. Trans-membrane (TM) domain and stress antifung (SA) domain were found in OsTPP8 and OsTPP11, respectively (Fig. 3b).

## 3-D Structure Analysis

3-D structure gives a few important residues that are associated with biological functions or desired interests (Bordoli and Schwede 2012). Therefore, we identified 3-D model of

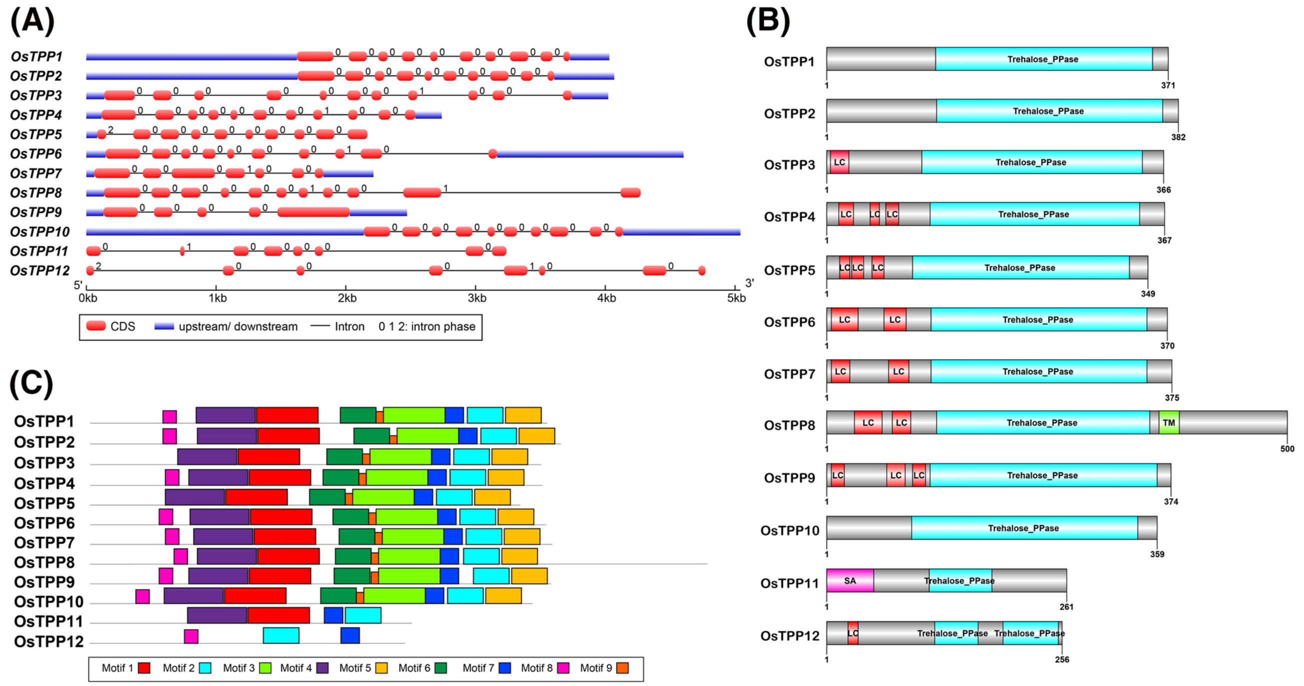


**Fig. 3** Schematic representation of the exon–intron arrangements, motif and domains of *OsTPP*s. **a** Exon–intron arrangements of *OsTPP*s. Red boxes represent the exons, and the black lines represent the introns. Un-translated regions (UTRs) are indicated by blue boxes, and the digits (0, 1 and 2) indicate the intron phase. **b** OsTPPs with domains drawn by DOG illustrator. Domains were identified using Pfam and SMART databases following the default parameters. *Trehalose_PPase* TPP domain, *SA* Stress Antifung, *TM* transmembrane region, *LC* Low complexity region. **c** Nine conserved motifs of OsTPPs. Motifs were identified using the online MEME program. Different colored boxes represent different motifs

OsTPPs from SWISS-MODEL and found three conserved residues within 4 A°, which acted as ligands (Waterhouse et al. 2018). Those ligands interacted between chain A and magnesium ion ($Mg^{2+}$), which revealed that OsTPPs have specific catalytic functions. It has been reported that TaTPP, ZmTPP, AtTPP, ScTPP, CaTPP and EcTPP acted as catalytic enzymes and all of them were 80% similar to each other (Acosta-Pérez et al. 2020). 3-D structure was analyzed with template "5gvx.1.A" for all of the OsTPPs. Predicted 3-D structures covering N-terminus and C-terminus regions of 12 OsTPPs proteins were identified by SWISS-MODEL (Fig. 4a). Secondary structure elements of protein sequences were calculated by SOPMA (Supplementary Table 5). Each of the OsTPP was composed of 32.95%–48.44% α helix, 14.05%–19.92% extended strand, 6.69%–9.20% β turn and 25.78%–41.88% random coil (Supplementary Table 5). In C-terminus, coiled coil-like structure was observed in first ten OsTPPs (OsTPP1 to OsTPP10) (Fig. 4a). Among 12 OsTPPs, eleven of them showed Magnesium ion ($Mg^{2+}$) ligands, but not OsTPP12 (Fig. 4a). OsTPP1 to OsTPP10 showed higher conserved structure than OsTPP11 and OsTPP12 (Fig. 4a). We used SWISS-MODEL assessment and Rampage server to validate OsTPP structures (Fig. 4B and Supplementary Table 6). The generated Ramachandran plot occupied with an average of 92.88% favored region, 5.97% allowed region and 1.16% outer region (Supplementary Table 6). Average sequence identity was 34.57%, while sequence similarity was 34.57%, which covered total 68% of query sequence found by X-ray Method in 2.6 A° (Supplementary Table 6).

The Protein–Ligand Interaction Pipeline (PLIP) confirmed the ligand interaction between chain A and $Mg^{2+}$, and that residue site was found to be highly conserved. We further analyzed these conserved residues in all OsTPP protein sequence alignments except OsTPP12 and found that these had aspartic acid (D/Asp) which is conserved in motif 4 and motif 2 (Fig. 5a). The side chains of the catalytic triads were displayed in enlarged form containing the residues of OsTPP7 for better visual understanding (Fig. 5b). To validate the modeling done by SWISS-MODEL, we used GASS-WEB (https://gass.unifei.edu.br/), which found more than 90% of the active or catalytic sites correctly (Izidoro et al. 2015) and was also experimentally proven for finding the catalytic site of the enzyme 2GCT (Gamma-Chymotrypsin A) (Moraes et al. 2017). Conserved residues found by GASS-WEB were slightly similar to the binding sites found by SWISS-MODEL (Supplementary Table 7). Interestingly, we observed the presence of $Mg^{2+}$ ligand along with their binding sites for different regions of OsTPP12, which we could not find in SWISS-MODEL (Supplementary Table 7).

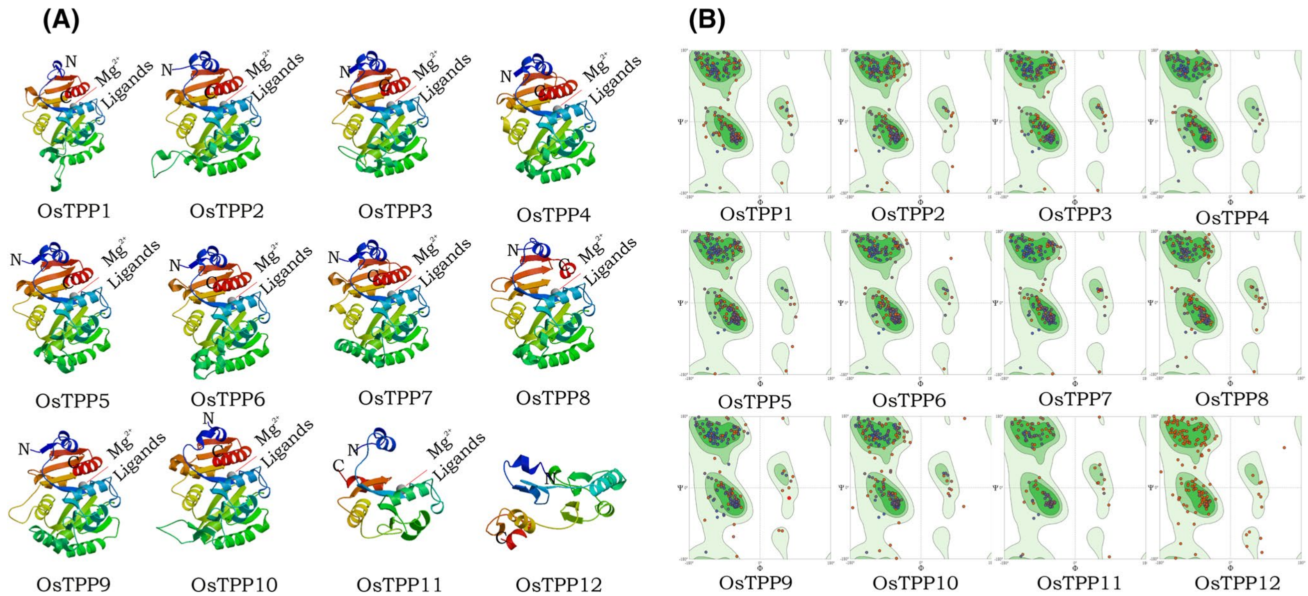



**Fig. 4** Schematic representation of 3-D protein structures and the Ramachandran plot for OsTPPs. **a** Predicted 3-D protein structures of OsTPPs. N, N-terminus; C, C-terminus; $Mg^{2+}$, magnesium ion. **b** The Ramachandran plot for OsTPPs

## *Cis*-Regulatory Element Analysis

CREs are important components of genetic regulatory network which control morphological changes in plants (Jiang and Rausher 2018; Ijaz 2020). Therefore, we analyzed 2 kb promoter regions immediately upstream of start codon (ATG) of all OsTPP genes using PlantCARE web tools and found a total of 83 CREs with a frequency of 1,497. The sequence lengths of these *cis*-elements were varied from 4 to 13 (Supplementary Fig. 1a ). CREs with a sequence length of 9 were found in the highest frequency (frequency 21), whereas CREs with a sequence length of 13 were found in the lowest frequency (frequency 1) (Supplementary Fig. 1a). The highest frequency in positive strand was 59, which was positioned between 101 and 200 bp upstream of ATG, and the lowest was 28, positioned between 501 and 600 bp upstream of ATG (Supplementary Fig. 1b). In case of negative strand, the highest frequency was 52, positioned between 1201 bp and 1300 bp, and 1701 bp and 1800 bp upstream of ATG, and the lowest was 18, positioned between 201 and 300 bp upstream of ATG (Supplementary Fig. 1b). We grouped these CREs into five groups (Group A to E) according to their known functions (Supplementary Fig. 1c and Supplementary Table 8).

Group A contained the core *cis*-elements, TATA-box and CAAT-box, which covered 39% of total CREs (Supplementary Fig. 1c and Supplementary Table 8). TATA-box (including TATA and AT ~ TATA-box) is a core promoter element generally present in around − 30 of transcription start site, and CAAT-box is a type of promoter which can influence transcription start site selection (Grace et al. 2004; Ijaz 2020). Generally TATA-box and CAAT-box are located 25–30 bp and ~ 75 bps upstream of the transcription start site, respectively (Godbey 2014). Both TATA-box and CAAT-box were found in all promoters in a wide range. TATA-box having a frequency of 245 covered 42% of CREs, and CAAT-box having a frequency of 342 covered 58% of CREs within the group. All core CREs in the promoters of OsTPPs were visualized in Supplementary Fig. 2a.

Group B contained total 47 stress-related CREs (SrCREs) covering 34% of all CREs found (Supplementary Fig. 1c and Supplementary Table 8). Each gene contained at least 14 SrCREs. Among them, only MYC was found in all of the gene promoters covering 9% of total SrCREs. Eleven gene promoters, except *OsTPP1*, contained G-box, a CRE involved in light responsiveness, which covered 8% of total SrCREs. ARE, STRE, Unnamed__1, W-box and WRE3 were found in ten OsTPP promoters. Among those, Unnamed__1 covered 11% and STRE covered 8% of total SrCREs. All SrCREs in the promoters of OsTPPs were visualized in Supplementary Fig. 2b.

Group C contained total 13 CREs related to cellular development (CdCREs) covered 18% of total CREs (Supplementary Fig. 1c and Supplementary Table 8). Each promoter contained 3–5 CdCREs in common. Among them, only Unnamed__4 that may function as temporal and spatial specificity of DYT1 (Anther) expression, located in all of the gene promoters contributed 83% of total CdCREs (Zhou et al. 2017). Nine gene promoters were found to have AAGAA-motif involved in seed specific expression that contributed 6% of total CdCREs (Zhou et al. 2019). All CdCREs

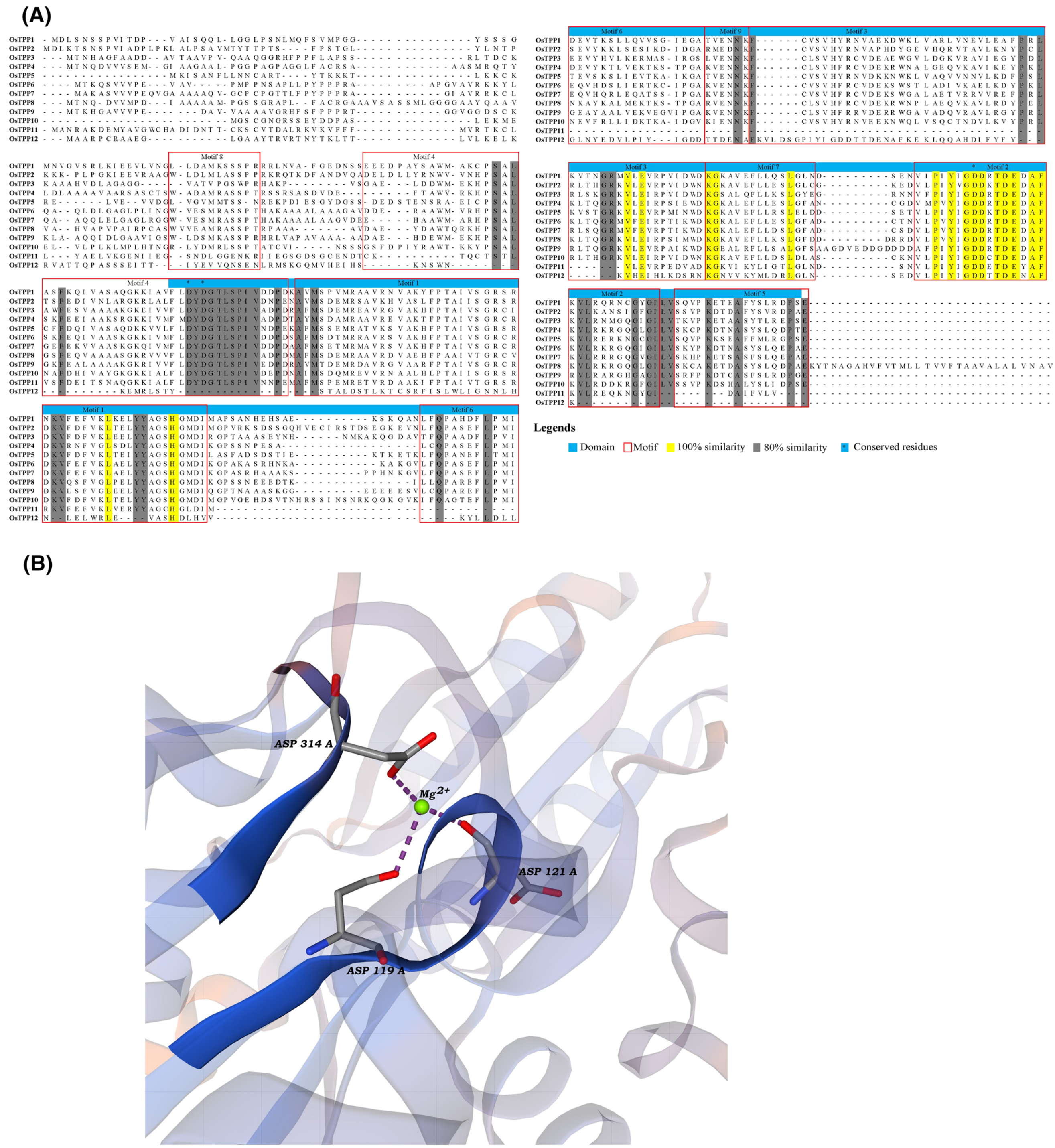



**Fig. 5** Schematic representation of protein alignment of all OsTPPs and the side chains of the catalytic triads of OsTPP7. **a** Protein alignments of all OsTPPs. **b** The side chains of the catalytic triads of OsTPP7

found in the promoters of OsTPPs were visualized in Supplementary Fig. 2c.

Group D contained total 17 CREs related to hormonal regulations (HrCREs) covered 8% of total CREs (Supplementary Fig. 1c and Supplementary Table 8). Each promoter contained 4–12 HrCREs. Eleven promoters contained ABRE composed of 37% of total HrCREs, which might function in response to ABA. O2-site HrCRE covering 10% of total CREs was found in ten OsTPP promoters, involved in zein metabolism regulation. Nine OsTPP promoters contained CGTCA-motif (16%), a HrCRE involved in the MeJA-responsiveness. Eight promoters contained TGA-element

(8%) that might function as auxin-responsive element. Seven genes were found to have ERE (10%), an ethylene-responsive element (Wei et al. 2019). Five genes had GARE-motif (5%), a gibberellin-responsive element; four genes had P-box (4%), a gibberellin-responsive element; three genes had TATC-box (2%) involved in gibberellin-responsiveness; three promoters had TCA-element (4%), an HrCRE involved in salicylic acid responsiveness. All HrCREs found in the promoters of OsTPPs were visualized in Supplementary Fig. 2d.

Group E contained four unknown CREs covering 1% of total CREs (Supplementary Fig. 1c and Supplementary Table 8). Seven OsTPP promoters contained Unnamed__2, which might act as an antisense transcript (Ash et al. 2013). Two promoters contained Unnamed__6; one promoter contained CTAG-motif and Unnamed__16. Unnamed__16 was found to be involved in sugar transporter family genes (Jiu et al. 2018). All CREs with unknown functions were visualized in Supplementary Fig. 2e.

### Expression Analysis

To investigate expression patterns of OsTPPs, we collected mRNA expression data from Genevestigator. The data were divided into 6 groups: Group 1 for callus, Group 2 for primary cell (cell culture / primary cell, panicle cell, spikelet cell, floret cell, pistil cell, ovary cell, ovule cell, embryo sac cell, egg cell, caryopsis cell, embryo cell and zygote), Group 3 for seedling (seedling, coleoptile, leaf, root, radicle, maturation zone, radicle tip, elongation zone), Group 4 for inflorescence (inflorescence, panicle, spikelet, floret, stamen, anther, pistil, caryopsis, embryo, endosperm), Group 5 for shoot (shoot, leaf, blade (lamina), flag leaf), and Group 6 for rhizome (rhizome, roots, nodal root, crown root, root tip, unspecified root type, root tip1). The results of 12 OsTPPs indicated that the expressions were significant for all except *OsTPP3* (from MeV t-test). By doing hierarchical clustering in MeV, OsTPPs were classified into 4 classes: class I (*OsTPP4*, *OsTPP7* and *OsTPP8*), class II (*OsTPP1*, *OsTPP6* and *OsTPP9*), class III (*OsTPP2*, *OsTPP5* and *OsTPP10*) and class IV (*OsTPP3*, *OsTPP11* and *OsTPP12*) (Fig. 6a). Class I and class II genes were expressed in most of the organs compared with class III and class IV genes. In class I, *OsTPP7* showed the highest expression in contrast to all other OsTPPs (Fig. 6a). In class III, only *OsTPP2* was expressed highly in most of the organs, and *OsTPP5* was expressed moderately in stamen and anther (Fig. 6a). In class IV, only *OsTPP3* was expressed higher in caryopsis, endosperm and inflorescence. The rest of the genes did not show higher expressions (Fig. 6a).

In developmental stages, *OsTPP1, OsTPP2, OsTPP6, OsTPP7* and *OsTPP9* were found to be expressed in all the stages. Particularly, *OsTPP3, OsTPP4, OsTPP5* and *OsTPP10* were highly expressed in flowering stage (Supplementary Fig. 3a). Gene Distance Matrix identified that some genes had similar expressions, such as *OsTPP1* with *OsTPP2*, *OsTPP6* and *OsTPP9*; *OsTPP2* with *OsTPP5, OsTPP6, OsTPP9* and *OsTPP10*; *OsTPP4* with *OsTPP6, OsTPP7, OsTPP8* and *OsTPP9*; *OsTPP5* with *OsTPP10; OsTPP6* with *OsTPP7* and *OsTPP9; OsTPP8* with *OsTPP9; OsTPP11* with *OsTPP12* (Supplementary Fig. 3b).

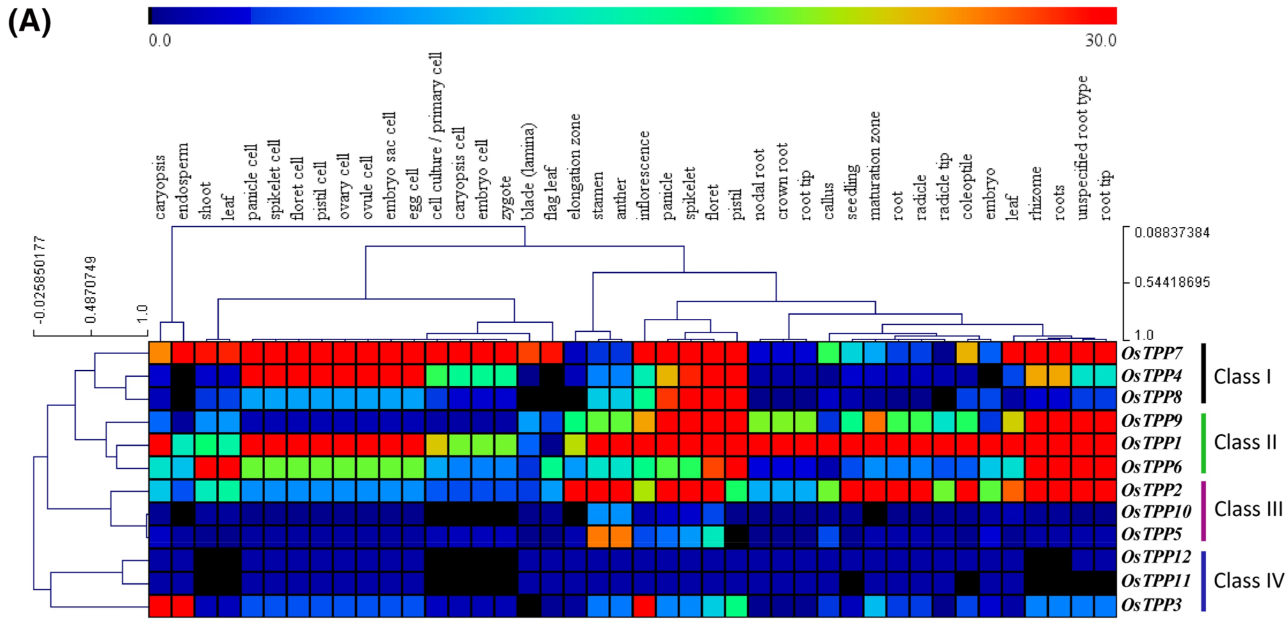
To examine responses of OsTPPs to abiotic stresses, we analyzed rice plants treated with 150 mM salt, 20% PEG-6000 (drought) and 8 °C cold stresses by qRT-PCR (Fig. 6b). Two genes, named OsTPP11 and OsTPP12, did not show any expression in qRT-PCR. Most of the remaining genes showed values less than two in log2 fold. *OsTPP2*, *OsTPP3*, *OsTPP8* and *OsTPP9* showed continuous upregulated expression in contrast to other genes in all three abiotic stress conditions, where *OsTPP8* and *OsTPP9* were highly upregulated (Fig. 6b). Downregulated expressions of *OsTPP1*, and *OsTPP4* to *OsTPP7* were found mainly for salt stress (Fig. 6b). Interestingly, all genes except *OsTPP1* showed variable ranges of upregulated expressions in response to drought and cold stresses (Fig. 6b), indicating that OsTPPs are more responsive to drought and cold compared to salt stress.

## Discussion

TPPs have been reported variably in several studies from rice and other species (Shima et al. 2007; Ge et al. 2008; Henry et al. 2014; Wang et al. 2019). Here, we identified total 12 OsTPPs using in silico approaches. We analyzed rice TPPs and other TPPs in the 11 species and interestingly, the number of OsTPPs was found second highest after soybean. *OsTPP1, OsTPP2, OsTPP6* and *OsTPP7* consisted of 5, 3, 2 and 3 splicing variants, respectively (Table 1). Three pairs of duplicated genes were also found (Fig. 1). From the chromosomal distribution, we identified that all the duplications were inter-chromosomal in nature. No tandem duplication was found for any OsTPPs.

Multiple alignments of OsTPP sequences were done by Muscle (codon) to identify the similar regions with minimum gaps (Fig. 5a). These similarities generally refer to functional equivalence and evolutionary relationships between different proteins. We analyzed Trehalose_PPase domain with conserved motif in multiple sequences and found that the motifs were highly conserved in N-terminus region (Fig. 5a). Among the identified nine motifs, motif 2 and motif 7 were found to be conserved in all OsTPPs. Nearly all the motifs, except motif 8, were highly conserved in first ten OsTPPs (OsTPP1 to OsTPP10). From our analyses, we found a short domain for OsTPP11 and a truncated domain for OsTPP12. OsTPP11 and OsTPP12

**(A)**




**(B)**

| | 150 mM Salt | | | | 20% PEG | | | | 8 °C Cold | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 3 h | 6 h | 12 h | 30 h | 3 h | 6 h | 12 h | 30 h | 3 h | 6 h | 12 h | 30 h |
| *OsTPP1* | 0.55 | -0.28 | 0.79 | 1.41 | 1.52 | 0.50 | -0.12 | 1.50 | 1.78 | 2.11 | 2.63 | 2.42 |
| *OsTPP2* | 0.40 | 1.44 | 1.81 | 2.00 | 2.25 | 1.87 | 1.32 | 2.54 | 0.94 | 1.39 | 1.74 | 2.07 |
| *OsTPP3* | 0.72 | 1.60 | 0.83 | 1.29 | 1.04 | 1.64 | 2.72 | 1.05 | 0.73 | 1.08 | 1.24 | 1.57 |
| *OsTPP4* | 0.44 | 1.01 | -0.46 | 1.06 | 0.69 | 0.93 | 1.30 | 1.26 | 0.26 | 0.07 | 0.16 | 0.78 |
| *OsTPP5* | 0.55 | 0.69 | -0.01 | 0.69 | 1.27 | 0.74 | 1.06 | 0.51 | 0.96 | 0.28 | 0.94 | 1.45 |
| *OsTPP6* | -0.57 | 0.71 | -0.77 | 0.67 | 0.68 | 1.20 | 0.88 | 1.43 | 0.33 | 0.48 | 0.90 | 1.31 |
| *OsTPP7* | 0.17 | 0.58 | -0.19 | 0.37 | 0.71 | 0.80 | 0.00 | 1.01 | 0.68 | 1.44 | 1.67 | 1.32 |
| *OsTPP8* | 1.01 | 3.18 | 1.13 | 1.93 | 2.94 | 2.19 | 3.87 | 1.62 | 2.63 | 1.57 | 1.86 | 3.33 |
| *OsTPP9* | 0.89 | 2.03 | 0.66 | 2.28 | 3.18 | 2.94 | 1.65 | 2.68 | 1.07 | 1.37 | 1.56 | 1.61 |
| *OsTPP10* | 0.60 | 0.58 | 0.34 | 0.39 | 0.14 | 0.37 | 1.22 | 0.62 | 0.23 | 0.30 | 0.54 | 1.41 |
| *OsTPP11* | | | | | | | | | | | | |
| *OsTPP12* | | | | | | | | | | | | |

**Fig. 6** Heat map of gene expression in organ specific and qRT-PCR expression in abiotic stress conditions. **a** Schematic representation of expression analysis of *OsTPPs*. Gene expression data was collected from Genevestigator (Os_mRNASeq_RICE). **b** Expression heat map of OsTPP genes based on corresponding log2 fold change in 150 mM NaCl, 20% PEG-6000 and 8 °C cold abiotic stress conditions at 3 h, 6 h, 12 h and 30 h. Values indicate relative expression levels with log2 fold in three biological replications from cDNA prepared from leaves. Heat map was done in Microsoft excel with default color scaling settings

both located far away from the cluster in radiation tree (Fig. 2b). Previously, *OsTPP11, OsTPP12* and *OsTPP13* (*LOC_Os06g23010*) were reported as putative pseudogenes (Ge et al. 2008). But we did not find the presence of any significant domain or motif in OsTPP13. Thus, we eliminated OsTPP13 from our study. Moreover, in two separate studies on pseudogene prediction (Thibaud-Nissen et al. 2009; Xie et al. 2019), we did not find any of these three genes predicted as pseudogenes. Further experimental studies would be necessary to validate this experimentally. Phylogenetic tree showed that monocots and dicots of all species were grouped into different sub-clades under two main clades. Radiation tree also indicated that OsTPP11 and OsTPP12 were moving far away from the cluster of all other TPPs (Fig. 2b).

Sub-cellular localization predictions of 12 OsTPPs were predicted in cytoplasm, nucleus, plasma membrane and chloroplast (Table 1). Different types of localizations were also observed in *Arabidopsis*. AtTPPD and AtTPPE were found to be localized in chloroplast. AtTPPA, AtTPPB, AtTPPC, AtTPPF, and AtTPPH fused to the cyan fluorescent protein (CFP) showed fluorescence in cytosol and nucleus. AtTPPG, AtTPPI, and AtTPPJ CFP fusion proteins showed localization in nucleus (Krasensky et al. 2014). From the multiple sequence alignment of OsTPP genes, we found that motifs were highly conserved in C-terminus. Due to lack of conserved N-terminus, their sub-cellular localizations were not same for all 12 OsTPPs. According to the 3-D structure analysis, first ten OsTPPs (OsTPP1 to OsTPP10) were more conserved than OsTPP11 and OsTPP12 (Fig. 4a). Besides this, first eleven OsTPPs (OsTPP1 to OsTPP11) showed $Mg^{2+}$ ligand-binding sites, except OsTPP12 in SWISS-MODEL. Interestingly, in GASS-WEB analysis, OsTPP12 showed $Mg^{2+}$ ligand-binding site (Supplementary Table 7). Magnesium has been reported to take part in catalytic activity by activating or inhibiting many enzymes (Cowan 1998; Bertini et al. 2007).

We analyzed 2 kb promoter regions of all OsTPPs to find CREs. Subsequently, we grouped the CREs into five main classes according to their known functions, namely core *cis*-elements, stress-related CREs, cellular development, hormonal regulations and CREs with unknown function (Supplementary Fig. 1c and Supplementary Table 8). We also visualized all CREs according to their groups for better observation (Supplementary Fig. 2). Frequency, specificity, position and combination of CREs, along with frequency and activity of transcription factors, occur in promoter control transcriptional regulation (Zou et al. 2011). In case of CRE frequency, stress-related CREs occur in higher percentage than CREs related to cellular development and hormonal regulations (Supplementary Fig. 4), clearly suggesting OsTPPs' involvement in stress response. ABA-responsive CREs, namely ABRE, At ~ ABRE, ABRE3a, and ABRE4, play significant roles in seed dormancy, stomatal closure, leaf senescence, biotic and abiotic stress response in plants (Leung and Giraudat 1998; Chak et al. 2000; Finkelstein et al. 2002; Himmelbach et al. 2003; Yamaguchi-Shinozaki and Shinozaki 2005). ABRE, ABRE3a and ABRE4 having the common ACGT core sequence are generally recognized by plant bZIP transcription factor, which involves in stress response and hormonal regulation (Guiltinan et al. 1990; Hattori et al. 1995; Hobo et al. 1999; Choi et al. 2000; Uno et al. 2000). ABRE3a and ABRE4 CREs were found in *OsTPP3*, *OsTPP4*, *OsTPP6*, *OsTPP8*, *OsTPP11*, and *OsTPP12* promoters. It has been reported that multiple ABREs or their combinations may act as a CE (Coupling Elements) to form ABRC (ABA-responsive complex) (Marcotte et al. 1989; Shen and Ho 1995; Shen et al. 1996; Hobo et al. 1999; Narusaka et al. 2003). Unnamed__1 CRE was found in ten OsTPP promoters except *OsTPP1* and *OsTPP10*. Unnamed__1 is an ABRE-like CRE, responsible for biotic and abiotic stress responses (Ishikawa et al. 2010). Myelocytomatosis oncogenes (MYC) CRE has been reported as developmental, hormonal and abiotic stress-related CRE, identified in different MYC-related gene promoters (Chen et al. 2019). Tran et al. (2004) reported MYC as a drought-responsive CRE in *Arabidopsis*. We found MYC in all OsTPP promoters suggesting critical involvement of OsTPPs in developmental, hormonal and stress responses, particularly in drought stress.

W-box CRE was reported to modulate gene expression during stress (Dhatterwal et al. 2019), which was found in ten OsTPP promoters except *OsTPP3* and *OsTPP5*. WRE3 and WUN-motif are recognized as wound-responsive elements (Singh et al. 2019; Wu et al. 2019). It has also been reported that WRE3 acts as a low-temperature responsive CRE (Singh et al. 2019). In our study, WRE3 was found in ten OsTPP promoters except *OsTPP1* and *OsTPP2*, and WUN-motif was found in *OsTPP3*, *OsTPP4*, *OsTPP7*, *OsTPP9* and *OsTPP11* promoters, indicating their possible role in wound response and cold stress. Unnamed__8 CRE (also known as Unnamed__10, Unnamed__12 and Unnamed__14) was found in *OsTPP3* and *OsTPP5* promoters. Unnamed__8 CRE acts as an Opaque-2 (O2) target site in maize, where O2 regulates gene expression in developing endosperm (Schmidt et al. 1992; Izawa et al. 1993; Neto et al., 1995). Plant_AP-2-like CRE was found in *OsTPP8*, which was also found in promoters of starch biosynthesis core genes of maize along with ABA-responsive CREs (Huang et al. 2016; Qu et al. 2019). Unnamed__4 CRE might regulate tissue-specific expression in plants, found in all OsTPPs (Zhou et al. 2017). AAGAA-motif might play an important role in early maturation, found in *OsTPP1*, *3*, *4*, *5*, *7*, *9*, *10*, and *11* (Kim 2013). Phytohormone-responsive CREs (auxin, MeJA, gibberellin, ABA, salicylic acid) were found in all OsTPP promoters, which could possibly assemble different pathways of signal transduction against stress response (Wolters and Jürgens 2009).

Data derived from Genevestigator (mRNA expression) were used for analyzing functional diversity of OsTPPs in different organs, but there were no expression data found for *OsTPP11* and *OsTPP12*. Thus, considering those ten *OsTPP*s, higher expressions of several genes were found in inflorescence and root (Fig. 6a). *OsTPP1*, *2*, *6*, *7*, and *9* were predominantly expressed in root, indicating that they could play a pivotal role in root physiology. *OsTPP7* enhanced anaerobic germination in rice by increasing coleoptile length, although their GUS assay using *OsTPP7* promoter showed expression in young heterotrophic tissues, embryo, coleoptile, root, aleurone and scutellum, independent of oxygen availability with relatively lower GUS activity in coleoptile (Kretzschmar et al. 2015). Our expression analysis also found *OsTPP7* expression in maximum number of tissues with higher expression pattern along with *OsTPP1* compared to others (Fig. 6a). Similar tissue-specific expression analysis was not reported for *OsTPP1*, but up-regulation of *OsTPP1* expression was found under abiotic stresses (Ge et al. 2008). In Zhonghua 11, a *japonica* rice variety, *OsTPP3* expression was found in roots, stems, stem nodes, seedlings, young panicles, leaves, and immature seeds (Jiang et al. 2019). Our analysis found significant *OsTPP3* expressions in inflorescence and seed. Interestingly, its duplicated gene *OsTPP9* (Fig. 1 and Supplementary

Table 3) was expressed mainly in spikelets, panicle and roots (Fig. 6a). These may suggest genotype-specific expression of certain OsTPPs. In addition, our qRT-PCR revealed remarkable altered expressions of OsTPPs to abiotic stress (Fig. 6b). For instance, *OsTPP2*, *OsTPP3*, *OsTPP8* and *OsTPP9* showed upregulated expression to salt, drought and cold treatments. *OsTPP1*, *OsTPP4*, *OsTTP5*, *OsTPP6*, and *OsTPP7* were down-regulated for salt stress. This clearly suggests their important role in abiotic stress tolerance.

This in silico analyses systematically characterized OsTPP family genes using different bioinformatics tools to explore their important role in growth, development and stress responses. In this study, we analyzed the gene structure, exon-intron distributions, chromosomal distribution, conserved domain, conserved motif, 3-D structure modeling and phylogenetic tree for evolutionary divergence. We also predict the biological functions on the basis of CREs and performed expression analysis which may serve as an important source for the future studies.

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**Author Contributions** JJ conceived and designed the experiments. MR, MR, and JE performed the experiments and analyzed the data. MR, MR, JE, and JJ wrote the paper.

## Compliance with Ethical Standards

**Conflict of interest** The authors declare no conflict of interest.

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